

Olive Chakraborty

📍 Paris ✉ olivebpilani@gmail.com ☎ 0769889478 in olive-chakraborty

Summary

Cryptographer with over 8 years of experience specializing in the design, cryptanalysis, and practical implementation of Post-Quantum Cryptography (PQC) and Fully Homomorphic Encryption (FHE). Proven track record in the NIST PQC Standardization process as the designer of the CFPKM key encapsulation mechanism. Extensive background in lattice-based cryptography, multivariate cryptography, algebraic cryptanalysis, and building privacy-preserving systems for machine learning and biometrics. Adept at bridging complex mathematical foundations with real-world performance constraints for secure, decentralized environments.

Skills

Cryptography and Engineering

Core Expertise: Post-Quantum Cryptography, Fully Homomorphic Encryption, Homomorphic Signatures, Threshold Cryptography, and Federated Learning.

Languages and Tools: C, C++, Python, Magma, SAGE, $\text{\LaTeX}$

Technical Leadership

Project Leadership: Work Package Lead for EU SafeTravellers; Contributor to EU Prividema.

Academic Service: Organizer of the Post Quantum Algebraic Cryptography Workshop, IHP Paris, 2024.

Education

Ph.D. in Computer Science *Sorbonne Université, Paris, France* 2017 – 2020

- **Thesis:** Design and Cryptanalysis in Quantum Safe Cryptography
- **Supervisor:** Jean-Charles Faugère & Ludovic Perret

M.Sc. in Information Security *Royal Holloway, University of London, UK* 2015 – 2016

- **Thesis:** Hardness of Learning With errors with Limited samples.
- **Supervisor:** Martin Albrecht (Information Security Group)
- Grade : First Class Honours with Distinction

M.Sc. in Mathematics *BITS Pilani, Pilani Campus, India* 2010 – 2015

- **Thesis:** Topological and Geometrical approximation of Singular Curves in 3D
- **Supervisor:** Marc Pouget (Team Gamble, INRIA Nancy)

B.Engg. in Computer Science *BITS Pilani, Pilani Campus, India* 2010 - 2015

- **CGPA:** 8.5/10

Experience

Research Scientist *Palaiseau, FR*
CEA LIST Nov 2021 – Present

- Conduct fundamental and applied research in cryptography, focusing on FHE and its application areas in Machine Learning with collaboration with academic and industrial partners.
- Lead EU-funded research projects on privacy-preserving identity verification and secure biometric systems. (Work-package leader: EU SAFETRAVELLERS, Contributor: EU PRIVIDEMA)
- Actively participate in project planning, proposal writing, and dissemination of results through conferences and journals.
- Supervise PhD and Master's students, providing mentorship on research methodology, publications, and

project development.

Post-Doc
CEA LIST

Palaiseau, FR
Feb 2021 – Oct 2021

- Designed a privacy preserving Speaker Recognition system leveraging Fully Homomorphic Encryption.
- Implemented a hybrid protocol combining encrypted linear layers with plain non-linear activations to enable real-time performance under FHE constraints.
- Achieved practical recognition latency ($\leq 1s$) and high accuracy on benchmark datasets using the TFHE scheme and optimizations tailored to speaker embeddings. Current results allow for a speaker recognition over 10 encrypted voices within 0.1 sec.
- Demonstrated the viability of secure speaker verification in outsourced environments without exposing sensitive biometric or model data.

Ph.D.

Sorbonne Université (ex Univeristé Pierre et Marie Curie)

Paris, FR
Jun 2017 – Dec 2020

- Designed and implemented a new multivariate key exchange scheme (CFPKM), submitted to the NIST PQC Standardization process, 2017.
- Provided the first known cryptanalysis of the Extension Field Cancellation (EFC) cryptosystem, revealing structural weaknesses and significantly improving the efficiency of existing attacks.
- Investigated algebraic cryptanalysis techniques on multivariate cryptographic schemes, focusing on the hardness of solving structured polynomial systems using Gröbner basis.
- Introduced and analysed a new hardness assumption based on solving polynomial systems with noise (PoS-SoWN), extending the LWE paradigm to multivariate settings.
- Developed algorithms for noisy polynomial system solving and demonstrated parameter-based attack strategies using hybrid algebraic techniques.

Research Intern
INRIA

Nancy, FR
Jun 2014 – Dec 2014

- Developed a certified numerical algorithm to describe the topology and geometry of singular algebraic curves arising from the projection of 3D curve intersections.
- Designed and implemented a robust solver in SageMath to isolate solution boxes in $\mathbb{R}^4$, ensuring correctness and termination guarantees via epsilon inflation and regularity analysis.
- Evaluated against state-of-the-art algorithms (Plantinga-Vegtor, Lin-Yap) and proposed improvements for handling complex singularity structures in real algebraic geometry.

Teaching Experience

Visiting Lecturer, Course: *Advanced Cryptography for Master 2*

Telecom Paris, Palaiseau
2024

- Designed and delivered a full course module covering theoretical foundations and practical implementations of PQC and FHE.
- Developed lecture content, hands-on lab exercises, and evaluation materials, ensuring alignment with the latest research and standardization efforts (e.g., NIST PQC).
- Introduced students to advanced topics such as lattice-based cryptography, Ring-LWE, CKKS/BFV schemes, and real-world applications of FHE in secure computation.
- Supervised mini-projects encouraging critical thinking and applied research in emerging cryptographic areas.

Publications

Journal Articles

1. J. Dumezy, A. Alexandru, Y. Polyakov, P.E. Clet, O. Chakraborty and A. Boudguiga, “Evaluating Larger Lookup Tables using CKKS,” Cryptology ePrint Archive, 2025 and FHE.org 2026 (to be presented).
2. J. Dumezy, O. Chakraborty, P.E. Clet, A. Boudguiga, “HEGIDE: A MIMD Oblivious Processor for Private Function Evaluation over CKKS,” Cryptology ePrint Archive (to appear).

3. A. Bondarchuk, O. Chakraborty, R. Sirdey and G. Couteau, “Downlink (T)FHE ciphertext compression,” *Selected Areas in Cryptography* (2025).
4. O. Chakraborty, J.-C. Faugère, and L. Perret, “Cryptanalysis of the extension field cancellation cryptosystem,” *Designs, Codes and Cryptography*, vol. 89, no. 6, pp. 1335–1364, 2021.
5. O. Chakraborty, J.-C. Faugère, and L. Perret, “CFPKM: A key encapsulation mechanism based on solving system of non-linear multivariate polynomials, NIST Post Quantum Cryptography Candidate,” 2017.

Conference Proceedings & Posters

1. J. Paillet, O. Chakraborty and M. Checri, “Pure multi-key BGV implementation,” in *ICISSP*, 2024.
2. O. Chakraborty and A. Boudguiga, “A decentralized federated learning using reputation,” in *ICISSP*, 2024.
3. O. Chakraborty, A. Pedrouzo-Ulloa, A. Boudguiga, R. Sirdey, O. Stan, and M. Zuber, “Practical multi-key homomorphic encryption for more flexible and efficient secure federated average aggregation,” in *2023 IEEE International Conference on Cyber Security and Resilience (CSR)*, IEEE, 2023, pp. 612–617.
4. O. Chakraborty and M. Zuber, “Efficient and accurate homomorphic comparisons,” in *Proceedings of the 10th Workshop on Encrypted Computing & Applied Homomorphic Cryptography (WACH @ CCS)*, 2022, pp. 35–46.
5. A. Pedrouzo-Ulloa, O. Chakraborty, A. Boudguiga, R. Sirdey, O. Stan, and M. Zuber, “Practical multi-key homomorphic encryption for efficient secure federated average aggregation - Preliminary work” Poster presented at the 6th HomomorphicEncryption.org Standards Meeting.
6. O. Chakraborty and S. Rath, “Hybrid modelling of flat rolling process during hot rolling,” in *Materials science forum*, Trans Tech Publ, vol. 762, 2013, pp. 277–282.

Thesis

1. O. Chakraborty “Design and Cryptanalysis of Post-Quantum Cryptosystems”, PhD Thesis. 2020.
2. O. Chakraborty “Hardness of Learning With Errors with Limited number of samples”, Master Thesis (Information Security), at Royal Holloway. 2016
3. O. Chakraborty “Numerical Algorithm for Topological and Geometrical Description of Singular Curves”, Master Thesis (Mathematics), at INRIA Nancy. 2014

Supervision / Encadrant

PhD Students

2023-Present

1. Antonina Bondarchuk, (2023-present) (co-supervised with Renaud Sirdey)
2. Jules Dumezy, (2024-present) (co-supervised with Aymen Boudguiga)
3. Aimad Chabounia, (to start Feb, 2026) (co-supervised with Nesrine Kaaniche)

Master Students

1. Marc Renard, (2022)
2. Justine Paillet, (2023)
3. Antonina Bondarchuk (2023)
4. Alice Garand (2025)